# Yiming Zhang

*Curriculum Vitae*

Institute for Rock Magnetism
University of Minnesota, Twin Cities
Minneapolis, MN 55455
yiming-z@umn.edu
duseryiming@gmail.com
+1 (323) 810-2226
duserzym.github.io


## ACADEMIC APPOINTMENT

*Postdoctoral Associate*, Institute for Rock Magnetism, University of Minnesota, Twin Cities Sep 2024-

*Postdoctoral Scholar*, Earth and Planetary Science, University of California, Berkeley May-Sep 2024

## EDUCATION

*Ph.D.*, Earth and Planetary Science, University of California, Berkeley 2024

*B.A, cum laude.*, Geology, Occidental College 2019

## PUBLICATIONS
## (* indicates mentored student)

**Zhang, Y**., Swanson-Hysell, N.L., Domeier, M., Gallo, L., Sapienza, F., Osorio Afanador, D.A, (2026), Enriching paleogeographic reconstruction through the integration of Bayesian age models into site-based apparent polar wander paths. JGR Solid Earth. https://doi.org/10.1029/2026JB034561

**Zhang, Y**., Hodgin, E.B., Schmitz, M.D., Schwartz, D., Mohr, M.T., Crowley, J.L, Swanson-Hysell, N.L., (2026), Protracted high-grade metamorphism and slow cooling in the Adirondacks recalibrate Neoproterozoic global paleogeography. Tectonics. https://doi.org/10.1029/2025TC009333

Kahn, L.X.*, **Zhang, Y**., Finnegan, Seth., Hodgin, E.B., Swanson-Hysell, N.L., (2025), The stratigraphic record of the arrival of the Sacramento and San Joaquin Rivers to the California coast. Journal of Sedimentary Research. https://doi.org/10.2110/jsr.2024.153

Fuentes, A.J., Courtney-Davies, Liam., Flowers, R.M., **Zhang, Y**., and Swanson-Hysell, N.L., (2025), Evolution of iron formation to ore during Ediacaran to early Paleozoic tectonic stability. Earth and Planetary Science Letters. https://doi.org/10.1016/j.epsl.2025.119621

Swanson-Hysell, N.L., **Zhang, Y**., Macdonald, F.A., Koran, I., Tasistro-Hart, A., and Jay, A., (2025), Oman was on the northern margin of a wide late Tonian Mozambique Ocean. Geology. https://doi.org/10.1130/G53450.1

Wenk, HR., Kattemalavadi, A., **Zhang, Y**., Kennedy, E.R., Borkiewicz, O., (2025), Exploring microstructures and anisotropies of serpentinites. Contributions to Mineralogy and Petrology. https://doi.org/10.1007/s00410-025-02209-5

**Zhang, Y**., Anderson, N.S.*, Mohr, M.T., Nelson, L.L., Macdonald, F.A., Schmitz, M.D., Thurston, O.G., Guenthner, W.R., Karlstrom, K.E., and Swanson-Hysell, N.L., (2024), Paleomagnetism of the southwest Laurentia large igneous province and Cardenas Basalt: pulsed magmatism during rapid late Mesoproterozoic plate motion. JGR Solid Earth. https://doi.org/10.1029/2024JB029036

Hodgin., E.B., Swanson-Hysell, N.L., Kylander-Clark, A.R.C., Turner, A.C., Stolper, D.A., Ibarra, D.E., Schmitz, M.D., **Zhang, Y**., Fairchild, L.M., Fuentes, A.J., (2024), One billion years of stability in the North American Midcontinent following two-stage Grenvillian structural inversion. Tectonics. https://doi.org/10.1029/2024TC008415

**Zhang, Y**., Hodgin, E.B., Alemu, T., Pierce, J.*, Fuentes, A., and Swanson-Hysell, N.L., (2024), Tracking Rodinia into the Neoproterozoic: new paleomagnetic constraints from the Jacobsville Formation. Tectonics. https://doi.org/10.1029/2023TC007866

Mohr, M.T., Schmitz, M.D., Swanson-Hysell, N.L., Karlstrom, K.E., Macdonald, F.A., Holland, M.E., **Zhang, Y**., Anderson, N.*, (2024), High-Precision U-Pb geochronology links magmatism in the SW Laurentia Large Igneous Province and Midcontinent Rift. Geology. https://doi.org/10.1130/G51786.1

Sapienza, F., Gallo, L.S., **Zhang, Y**., Vaes, Bram., Domeier, Mathew., Swanson-Hysell, N.L., (2023), Quantitative Analysis of Paleomagnetic Sampling Strategies. JGR Solid Earth. https://doi.org/10.1029/2023jb027211

Gallo, L. C., Domeier, M., Sapienza, F., Swanson-Hysell, N. L., Vaes, B., **Zhang, Y**., Arnould, M., Eyster, A., Guřer, D., Király, Á., Robert, B., Rolf., T., Shephard, G., and van der Boon, A., (2023), Embracing uncertainty to resolve polar wander: a case study of Cenozoic North America. Geophysical Research Letters. https://doi.org/10.1029/2023GL103436

Slotznick, S.P., Swanson-Hysell, N.L., **Zhang, Y**., Clayton, K.E., Wellman, C.H., Tosca, N.J., and Strother, P.K., (2023), Reconstructing the paleoenvironment of an oxygenated Mesoproterozoic shoreline and its record of life. GSA Bulletin. https://doi.org/10.1130/B36634.1

Pierce, J.*, **Zhang, Y**., Hodgin, E.B., and Swanson-Hysell, N.L., (2022), Quantifying inclination shallowing and representing flattening uncertainty in sedimentary paleomagnetic poles. Geochemistry, Geophysics, Geosystems. https://doi.org/10.1029/2022GC010682

Rose, I., **Zhang, Y**., and Swanson-Hysell, N.L., (2022), Bayesian paleomagnetic Euler pole inversion for paleogeographic reconstruction and analysis. JGR: Solid Earth. https://doi.org/10.1029/2021jb023890

**Zhang, Y**., Swanson-Hysell, N.L., Avery, M.S., Fu, R.R., (2022), High geomagnetic field intensity recorded by anorthosite xenoliths requires a strongly powered late Mesoproterozoic geodynamo. PNAS. https://doi.org/10.1073/pnas.2202875119

Hodgin, E.B., Swanson-Hysell, N.L., DeGraff, J.M., Kylander-Clark, A.R.C., Schmitz, M.D., Turner, A.C., **Zhang, Y**., Stolper, D.A., (2022), Final inversion of the Midcontinent Rift during the Rigolet Phase of the Grenvillian Orogeny. Geology. https://doi.org/10.1130/G49439.1

Cromwell, G., **Zhang, Y**., (2021), New paleointensity data from Aniakchak volcano, Alaska, USA. Geochemistry, Geophysics, Geosystems. https://doi.org/10.1029/2021gc010032

**Zhang, Y**., Swanson-Hysell, N.L., Schmitz, M.D., Miller Jr., J.D., and Avery, M.S., (2021), Synchronous emplacement of the anorthosite xenolith-bearing Beaver River diabase and one of the largest lava flows on Earth. Geochemistry, Geophysics, Geosystems. https://doi.org/10.1029/2021GC009909

Swanson-Hysell, N.L., Avery, M.S., **Zhang, Y**., Hodgin, E.B., Sherwood, R.J., Apen, F.E., et al., (2021). The paleogeography of Laurentia in its early years: new constraints from the Paleoproterozoic East Central Minnesota batholith. Tectonics. `https://doi.org/10.1029/2021TC006751`

Swanson-Hysell, N.L., Hoaglund, S.A., Crowley, J.L., Schmitz, M.D., **Zhang, Y**., and Miller Jr., J.D., (2020), Rapid emplacement of massive Duluth Complex intrusions within the Midcontinent Rift. Geology. `https://doi.org/10.1130/g47873.1`

**Zhang, Y**., Pairing paleointensity results with coercivity spectra: providing support for selection criteria. *IRM Quarterly. Volume 30. Number 1.*

## TALKS

2026 GSA Annual Meeting October 11-14, 2026
*U-Pb joint inversion for thermal history reconstruction*

2026 GSA Annual Meeting October 11-14, 2026
*Cryogenian collision and stitching plutonism in the northwestern Arabian-Nubian Shield preceded final closure of the Mozambique Ocean*

Space Physics Seminar, University of Minnesota Twin Cities (invited) April 14, 2026
*The Geodynamo Through Time: core paradoxes, paleomagnetic insights, and open questions*

Geology Seminar, Occidental College (invited) November 4, 2025
*Deep time, moving continents, and a snowball Earth*

Division Seminar, Caltech (invited) November 3, 2025
*Assembling supercontinent Rodinia and tracking its development in the late Proterozoic*

2025 GSA Annual Meeting October 19-22, 2025
*Recalibrating the history of metamorphism and exhumation in the Adirondack Highlands leads to a major revision of Neoproterozoic global paleogeography*

2025 GSA Annual Meeting (invited) October 19-22, 2025
*Paleogeographic Evolution of Oman and Egyptian Terranes of the Arabian-Nubian Shield through the Neoproterozoic*

Earth & Environmental Sciences Department seminar, UMN Duluth April 8, 2025
*Anorthosites in northern Minnesota: 100 years of study of their 1-billion-year-old history*

Scripps Institute for Oceanography October 7, 2024
*Reconstructing late Proterozoic magmatism, geomagnetic field behavior, and paleogeography by pairing paleomagnetism with state-of-the-art geochronology*

Hard Rock Lunch seminar, University of Minnesota Twin Cities October 2, 2024
*Reconstructing Neoproterozoic paleogeography with metamorphic rocks*

UC Berkeley EPS exit seminar April 25, 2024
*Reconstructing late Proterozoic magmatism, geomagnetic field behavior, and paleogeography using tiny magnets in rocks*

2023 AGU Fall Meeting Dec 5 2023
*Constraining the position of Arabian-Nubian arc terranes in the lead-up to Snowball Earth glaciation: new constraints from Tonian dikes of Oman*

Institute of Geology and Geophysics seminar, Chinese Academy of Science (*invited)* *July 5 2023*
*New approaches to APWP synthesis and incorporating uncertainties in sedimentary records*

2023 IRM conference (*invited)* *June 5-June 8 2023*
*New approaches to apparent polar wander path development constrain the late Mesoproterozoic assembly of Rodinia*

2023 MagIC workshop (*invited)* *Feb 28-Mar 2 2023*
*New perspectives on Laurentia's Grenville Loop: tracking Rodinia across the Mesoproterozoic to Neoproterozoic boundary*

2022 AGU Fall Meeting Dec 12-16 2022
*Reconstructing the position of the supercontinent Rodinia in the early Neoproterozoic: new constraints from Laurentia's interior and the Grenville margin*

Beijing Paleomagnetism and Geochronology Laboratory (*invited, online)* *Sep 28 2022*
*High geomagnetic field intensity recorded by anorthosite xenoliths requires a strongly powered late Mesoproterozoic geodynamo*

Young CEED 21 Frontiers in quantitative paleogeography (*invited, online)* *Nov 14-20 2021*
*Bayesian_PEP_inversion: a Bayesian framework for integrating paleomagnetic and geochronologic data into apparent polar wander inversion*

Grand Canyon Supergroup Field Forum (*invited)* *April 9-19 2021*
*The rich paleomagnetic record of the Mesoproterozoic Midcontinent Rift and the Southwestern Laurentia Large Igneous Province*

Cloud Meeting on Paleomagnetism (*online)* *1/29/2021*
*Intense magmatic activity and a strong geomagnetic field—a study on the anorthosite xenoliths hosted in the Mesoproterozoic Midcontinent Rift diabase*

2020 AGU Conference (*online)* *12/15/2020*
*Recovering Mesoproterozoic geomagnetic field intensity using anorthosite xenoliths hosted in Midcontinent Rift diabase*

North Central GSA Conference, Duluth, MN (*online)* *05/18/2019*
*The rich paleomagnetic records of Proterozoic Midcontinent Rift intrusives: an updated synthesis with a new pole from the Beaver River diabase*

Institute for Rock Magnetism, University of Minnesota, Minneapolis, MN (*invited)* *01/09/2019*
*Paleomagnetism and rock magnetism on the Beaver River diabase and anorthosite xenoliths therein*

## TEACHING

Lecturer at the IRM summer school July 2026
*micromagnetic modeling & magnetic microscopy*
`https://duserzym.github.io/rock_magnetometry/`

Teaching assistant for lab sessions at the IRM summer school June 2024

Graduate student instructor (GSI) for EPS 101 Field Geology and Digital Mapping Fall 2023
*Advisor: Matthew Gleeson*

Reader for EPS 88 PyEarth: A Python Introduction to Earth Science — Spring 2023
*Advisor: Nicholas Swanson-Hysell*

GSI for EPS 101 Field Geology and Digital Mapping — Fall 2022
*Advisor: Nicholas Swanson-Hysell*

Reader for EPS 115 Stratigraphy and Earth History — Spring 2022
*Advisor: Eben Blake Hodgin*

GSI for EPS 101 Field Geology and Digital Mapping — Fall 2021
*Advisor: Nicholas Swanson-Hysell*

GSI for EPS 50 The Planet Earth — Spring 2021
*Advisor: Daniel Stolper*

Participant in the Graduate Remote Instruction Innovation Fellows Program — Winter 2020

GSI for EPS 101 Field Geology and Digital Mapping — Fall 2020
*Advisor: Nicholas Swanson-Hysell*

Completion of UC Berkeley GSI Conference training — Jan 2020

Completion of required Online Course: Professional Standards and Ethics for GSIs — Fall 2019

Completion of required Pedagogy Course EPS 375 — Fall 2019

## ORIGINAL GEOLOGICAL FIELD WORK

Southern Sweden, [1 week] — 2026
*Reconstructing the paleogeographic position of Baltica in the Neoproterozoic*

Central Eastern Desert, Egypt [2 weeks] — 2025
*Reconstructing the Neoproterozoic paleogeography of the Arabian-Nubian shield*

Sliderock Mountains, Montana [1 week] — 2024
*Extending site-based apparent polar wander path for North America into the Mesozoic with paleomagnetic data from the intrusive rocks of the Sliderock Mountain Volcanics*

Oman [5 weeks] — 2024, 2026
*Constraining the paleogeographic configuration of the Mozambique Ocean in the Neoproterozoic*

Grenville orogen [3 weeks] — 2022, 2023
*Pairing paleomagnetic data and thermochronology records to reevaluate the exhumation history of rocks of the Grenvillian Orogeny and recalibrate the Grenville Loop*

Death Valley, California; Grand Canyon, Arizona [4 weeks] — 2021
*Using paleomagnetism and geochronology to study the temporal and magmatic relationship between the ca. 1.1 Ga Southwestern Laurentia Large Igneous Province and the Midcontinent Rift*

Pikes Peak, Colorado [2 weeks] — 2020
*Using paleomagnetism and geochronology to study the emplacement history of Pikes Peak batholith and its temporal and magmatic associations with the Midcontinent Rift 1.1 billion years ago*

Midcontinent Rift, Lake Superior Region [11 weeks] 2019, 2020, 2021
*Reconstructing the behavior of the Mesoproterozoic geomagnetic field, continental motion during supercontinent assembly and the nature of ancient environments through paleomagnetic studies of the intrusive rocks and sediments of the 1.1 billion-year-old Midcontinent Rift*

Central Highland, Iceland [3 weeks] 2019
*Qualitatively and quantitatively measure the erosion rate of rofabards soil erosion in Central highlands, Iceland*

## AWARDS AND FUNDING

UMN Post Doctoral Association Outstanding Postdoc Award 2026
*Teaching and Mentoring*

IAGA Early Career Presentation Awards 2025

AGU GPE Post-doc grant ($755) 2024
*Enhancing site-based apparent polar wander paths with Bayesian stratigraphic age models*

UC Berkeley graduate student conference travel grant ($900) 2023
*2023 Geochronology Gordon Research Conference*

UC Berkeley Earth and Planetary Science George C. Louderback Award 2023

2023 AGeS3-Grad Geochronology Award ($8,865) 2023
*Dating the Grenville Loop using U-Pb apatite thermochronology*

2022 GSA Graduate Student Research Grant ($1,749) 2022
*Paleomagnetism and thermochronology of the Adirondack Mountains, Grenville Province*

UC Berkeley graduate student conference travel grant ($900) 2022
*2022 AGU Fall Meeting oral presentation*

Hearts to Humanity Eternal (H2H8) Programs ($10,000) 2022
*H2H8 Association Graduate Research Grant to Advance Humanity through Science*

U.S. Visiting Student Fellowships, Institute for Rock Magnetism 2019
*Paleomagnetism and rock magnetism study on Mesoproterozoic Beaver Bay Complex and anorthosite xenoliths therein ($500)*

ILSG Student Research Fund, Institute on Lake Superior Geology 2019
*To study the emplacement history of the Beaver River diabase and the anorthosite xenoliths therein using paleomagnetism ($500)*

Chevron-Xenel Gateway Fellowship, Berkeley International House ($5,000) 2019

John Parke Young Student Grant, Occidental College 2019
*Using UAV Data for Monitoring Modern Rofabard Soil Erosion in Central Highlands, Iceland ($3,500)*

Independent research, Occidental College 2018
*Pseudo-Thellier Paleointensity Measurement on R-N Geomagnetic Polarity Reversal Recorded by Mafic Lava Flows, Anahola, Kauai ($4,000)*

Independent research, Henry Luce Foundation, Nanjing University 2017

*Mapping of Ambient Ozone Pollution in China and the Assessment of Its Health Impact on Socio-Economy ($2,250)*

## SERVICE

Contributing developer to the open source PmagPy software project (`https://github.com/PmagPy/PmagPy`).

Contributing developer to the open source RockmagPy software project (`https://github.com/PmagPy/RockmagPy-notebooks`).

Lead developer for the RapidPy superconducting rock magnetometer automatic sample changing system control software (`https://duserzym.github.io/RAPID/`).

## IN THE NEWS

*GSA ConnectsCast (2026): Putting Numbers on Deep Time Using Zircon Crystals at Boise State's Isotope Geology Laboratory*
Featured interview on how high-precision U-Pb zircon geochronology helps reconstruct a billion-year-old Earth and establish the timing of major events in Earth's history.
`https://www.youtube.com/watch?v=JFVQO4tFayI`

*PNAS In This Issue:Evidence of strong Proterozoic magnetic field*
*`https://www.pnas.org/doi/10.1073/iti2922119`*

*EOS news: Swinging Strength of Earth's Magnetic Field Could Signal Inner Core Formation*
`https://eos.org/articles/swinging-strength-of-earths-magnetic-field-could-signal-inner-core-formation`

*UC Berkeley EPS department news: EPS Research on Earth's Ancient Magnetic Field Featured in EOS:*
`https://eps.berkeley.edu/news/eps-research-earths-ancient-magnetic-field-featured-eos`

*Diamonds Are a Paleomagnetist's Best Friend in EOS:*
`https://eos.org/articles/diamonds-are-a-paleomagnetists-best-friend`

## Reviewer for the following journals

*Earth and Planetary Science Letters*
*Geochemistry, Geophysics, Geosystems*
*GEOLOGY*
*Nature Communications Earth & Environment*
*Science Bulletin*
*Tectonics*
*Science Advances*

## Reviewer for the following panels

*NASA Emerging Worlds*
*National Science Foundation*

## MEMBERSHIPS

American Geophysical Union (AGU)
Geological Society of America (GSA)